

Space based laser is the ultimate objective of US-directed energy laser programme. It is proposed to be configured around 20MW hydrogen fluoride laser operating at 2.7µm. It is being designed to intercept intercontinental ballistic missiles and other strategic and tactical missiles. It is proposed to deploy the system in 800km to 1300km orbit. The expected target engagement range is 4000km to 12,000km.

Several directed energy laser systems based on solid-state and fibre lasers are being developed and tested for tactical mission needs ranging from ordnance neutralisation to anti-missile and anti-RAM applications.

Raytheon has developed and successfully tested a directed energy laser system called Laser Phalanx (Fig. 12) employing a 20kW industrial fibre laser. The system has been successfully demonstrated against a static mortar from a distance of 0.5 kilometres.

Raytheon has also successfully tested a ship-mounted solid-state laser weapon system (Fig. 13) to shoot four drones.

Laser-induced plasma channel weapons

A laser-induced plasma channel weapon is conceptually different from conventional high-energy laser weapons. It makes use of an ultra-short laser pulse to create a highly-conducting plasma channel between the laser and the intended target. It is designed to take out targets that conduct electricity better than air or ground surrounding these.

This conducting plasma follows the path of the laser and therefore can be directed to different targets by steering the laser beam. When the plasma comes in contact with a high-voltage source, a high-



Fig. 12: Laser Phalanx



Fig. 13: Laser weapon system

voltage current discharge travels down the plasma channel and then through the target to ground, thereby causing severe damage to the target—similar to what would have been caused had there been a lightning strike.

The concept behind formation of plasma channel is as follows: A pulsed laser of even a modest energy producing ultra-short laser pulses is capable of generating gigantic peak power of the order of tens of gigawatts. The electromagnetic field produced due to the intense laser beam rips off electrons from the air molecules, thereby ionising the surrounding air and creating plasma. For high-intensity laser pulses, air can act like a lens. The laser focuses on itself in air, confining light to a small-diameter filament.

Laser-induced plasma channel weapons behave like a lightning strike that always tries to follow

the path of least resistance while travelling from cloud to ground. Plasma channel conducts electricity much better than un-ionised air. If the plasma channel comes near a high-voltage source, electrical energy travels down the ionised conduit. When a laser-induced plasma channel weapon is used on a target such as a vehicular platform, a person or unexploded ordnance, high-voltage current discharges through the path of least resistance to ground, potentially disabling the vehicle or person, and initiating ordnance detonation.

Laser-induced plasma channel weapon hardware mainly comprises a laser capable of generating ultra-short pulses of the order of a few picoseconds and a power source to drive both laser and high-voltage discharge. But there are many technological challenges to be overcome. These include synchronising the laser with the high voltage, ruggedising the device to survive under extreme environmental conditions of an operational environment, and powering the system for extended periods of time.

Laser-induced plasma channel devices can have a variety of applications. These can be used to kill or incapacitate human targets through electric shock. These can also damage, disable and destroy electronic devices.

Laser-induced plasma channels may also be used for a variety of experiments, including study of lightning discharges, forcing lightning discharges to occur at safe time and place during thunderstorms, inducing thunderheads that deliver a precise lightning strike on to a ground target triggered by an airborne laser, and harvesting lightning energy for power generation by directing it to a terrestrial collection station. **EFY**